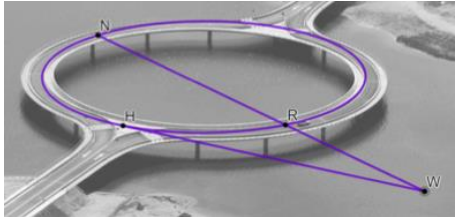


12.3 Solving Problems with Secant and Tangent Segments

Lesson Introduction

 Benchmark	MA.912.GR.6.1 Solve mathematical and real-world problems involving the length of a secant, tangent, segment, or chord in a given circle.		 Learning Target	I can solve mathematical and real-world problems that involve the lengths formed by the intersection of a secant and a tangent segment.
 Benchmark Coherence	Building On		Working Toward	
	N/A		N/A	
 Essential Questions	- What is the relationship of the lengths formed by the intersection of secant and tangent segments? - How can you use the intersection of secant and tangent segments to find the length of missing line segments?			
 Lesson Narrative	This lesson will introduce the students to secant and tangent segments. They will explore the relationship of intersecting secant and tangent lengths, and will work with a partner to write a formula that can be used to find the lengths of missing line segments. Students will be guided through a series of questions, including a question involving a drone flying over Laguna Garzon Bridge in Uruguay. They will conclude by trying questions on their own, including identifying and correcting an error in work.			
 Component Summary	12.3.1 Warm-Up (~5 min.)	Notice and Wonder with patchwork pattern (<i>SE p. 242</i>)	12.3.3 Guided Practice (10-15 min.)	Solving problems involving intersecting secant and tangent lengths (<i>SE p. 244</i>)
	12.3.2 Exploration (10-15 min.)	Discovering intersecting secant and tangent lengths (<i>SE p. 243</i>)	12.3.4 Your Turn! (10-15 min.)	Independent problem solving involving intersecting secant and tangent lengths (<i>SE p. 245</i>)
	12.3.5 Wrap-Up (~1 min.)	Students summarize their learning (<i>SE p. 246</i>)		
 Resources	Glossary		Materials	Additional Preparation
	<u>Key Terms:</u> N/A <u>Additional Glossary:</u> - Secant - Tangent		(Optional) Colored Pencils or Markers (Optional) Chart Paper	N/A

 Teacher Tips	Common Misconceptions - Students may be confused between the tangent ratio from trigonometry and the tangent of circles. - Students may confuse multiplying, squaring, and adding in the formula. - Students may confuse a secant line and a tangent line.	Things to Consider Emphasize the point of tangency, and discuss the intersection of the radius to the point of tangency as perpendicular. As you illustrate this theorem, be sure to include a triangle whose sides are the radius of the circle, a line of tangency perpendicular to it, and a segment from the center to a point on the tangent line other than the outer endpoint of the radius.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Take Turns In 12.3.2 Exploration students work with a partner. They take turns, whether it be explaining, justifying, agreeing or disagreeing, or asking clarifying questions. If they disagree, they are expected to support their case and listen to their partner's arguments. The first few times, consider demonstrating with a partner how the discussion is expected to go. Once students are familiar with these structures, less set-up will be necessary. While students are working, ask them to restate their responses more clearly or paraphrase what their partners said.	MA.K12.MTR.4.1: Discussion and Reflection 12.3.2 Exploration allows the students to communicate mathematical ideas, vocabulary, and methods effectively. 12.3.2 Exploration will develop students' ability to justify methods and compare their responses to the responses of their peers.
	 Differentiate Instruction	Consider illustrating the theorem on a separate piece of chart paper that you will leave hanging in the classroom as a reference for the students. Students can also co-construct Anchor Charts, either during 12.3.2 Exploration question 1 or the real-world question 3 during 12.3.3 Guided Practice. 
Lesson Notes:		

12.3.1 Warm-Up: Notice and Wonder

Component Overview: Students are shown a Seminole Native American tribal blanket and asked to write down things they notice and wonder about the patchwork on the blanket. The purpose is to make a mathematical task accessible for all students with these low-stakes questions.



12.3.1 Warm-Up: Notice and Wonder

1. The Seminole Native American tribe is known for their novel and colorful dress consisting of patchwork. The decorative technique of patchwork was developed by Seminole women before 1920. Patchwork was rapidly adopted to further embellish the already colorful clothing. As time went on, the designs became more intricate as the seamstresses became more skilled.



- a. What do you notice about the patchwork?
Answers will vary. They use a particular pattern for an entire row. Most of the patterns contain a quadrilateral.
- b. What do you wonder?
Answers will vary. I wonder how they come up with the patterns. Do they have special meaning?



Students can engage in several rounds of Notice and Wonder, so use this as an opportunity to reinforce the expectations and routine of a Notice and Wonder.

12.3.2 Exploration: Lengths of Intersecting Secant and Tangent Segments

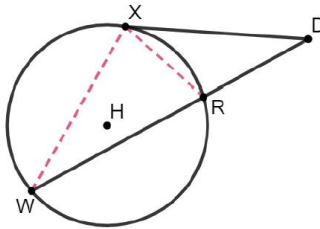
Component Overview: Students are working with intersecting secant and tangent lengths. The outcome of this section is for students to be led into finding the formula for intersecting secant and tangent lines on their own. Students will engage in MA.K12.MTR.4.1 in this collaborative component. Consider giving students space and opportunities to analyze errors and engage in math talk and collaboration.



12.3.2 Exploration: Lengths of Intersecting Secant and Tangent Segments

Work with your partner to complete the following.

- Circle H has points X , R , and W on the circle. Secant $\overline{WD}$ and tangent $\overline{XD}$ intersect at point D outside the circle. It is given that $\triangle DWX \sim \triangle DXR$, $XD = 15$, and $RD = 9$.



- Use the similar triangles to write a proportional relationship that can be used to determine the length of $\overline{RW}$.

$$\frac{XD}{RD} = \frac{WD}{XD}$$

$$\frac{15}{9} = \frac{RW + 9}{15}$$



Notice and Wonder

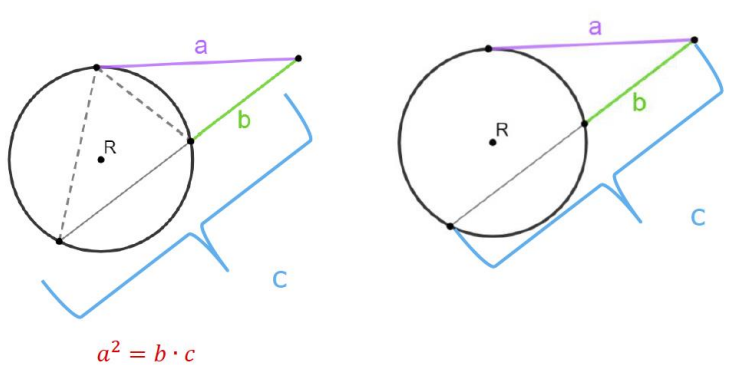
Consider giving students the opportunity to write down things they notice and wonder about the diagrams. This will allow students the chance to make observations before answering the questions.



Students need to pay close attention to the details of the line segment to accurately determine if the line segment is a tangent, secant, or chord before solving questions 1a-b. Encourage precision in responses.

12.3.2 Exploration: Lengths of Intersecting Secant and Tangent Segments (continued)

- b. Use the diagrams of circle R to write a formula that can be used to find the segment lengths of the intersecting secant and tangent lines.



If a secant and tangent segment are drawn to a circle from the same external point, then the square of the measure of the tangent segment is equal to the product of the measures of the secant segment and its external secant segment.

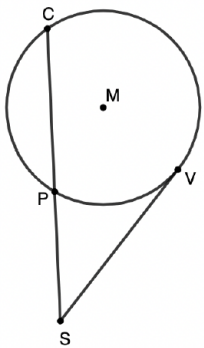
12.3.3 Guided Practice: Solving Problems Involving Intersecting Secant and Tangent Segments

Component Overview: Students are applying their conceptual understanding of intersecting secant and tangent lengths and the formula derived in 12.3.2 Exploration to find the length of line segments.



12.3.3 Guided Practice: Solving Problems Involving Intersecting Secant and Tangent Segments

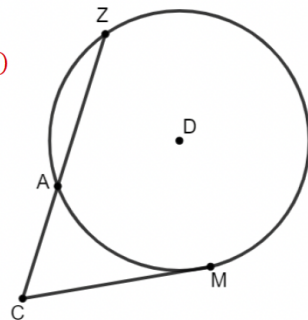
1. Circle M has points C , V , and P on the circle. Secant $\overline{CS}$ and tangent $\overline{VS}$ intersect at point S outside the circle. $CP = 4$ and $SP = 3$. What is the length of $\overline{SV}$? Round to the nearest thousandth.



$$\begin{aligned} SP(SP + CP) &= (SV)^2 \\ 3(7) &= (SV)^2 \\ SV &= 4.583 \end{aligned}$$

2. Circle D is shown with points Z , A , and M on the circle. Secant $\overline{CZ}$ and tangent $\overline{CM}$ intersect at point C outside the circle. $AZ = 16$, $AC = (x - 3)$, and $MC = (x + 3)$. What is the length of $\overline{CM}$?

$$\begin{aligned} (x + 3)^2 &= (x - 3)(x - 3 + 16) \\ x &= 12 \\ CM &= 15 \end{aligned}$$



Use informal observations from 12.3.2 Exploration to inform how to engage students with this activity.

- If student responses sound similar to the blue box in 12.3.2 Exploration, have students lead the class and apply their findings to question 1.
- If students complete 12.3.2 Exploration and struggle making a connection between secants and tangents, use question 1 as a model before students try question 2 in pairs.

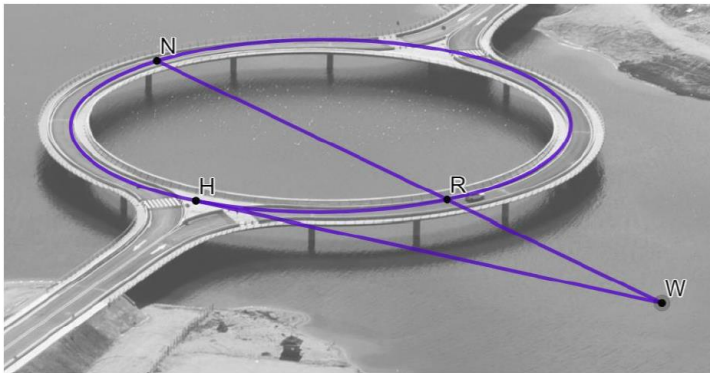


Think-Pair-Share

Allow students time to think about how they would set up the equation. Students can discuss with a partner before sharing with the class. This will ensure students are setting up the equation before solving.

12.3.3 Guided Practice: Solving Problems Involving Intersecting Secant and Tangent Segments (continued)

3. A drone flew over the Laguna Garzon Bridge in Uruguay and took pictures at three points on the bridge: N , R , and H . Tangent $\overline{HW}$ and secant $\overline{NW}$ intersect at point W off the bridge. If $HW = 280$ feet (ft) and $RW = 160$ ft, what is the length of $\overline{NW}$?



$$280^2 = 160(NW), \quad NW = 490 \text{ ft}$$



Allow students to engage in meaningful dialogue related to this real-life situation. Consider using a Notice and Wonder instructional routine. After students have had a chance to think about their responses, ask several students to share things they notice and things they wondered about this question.

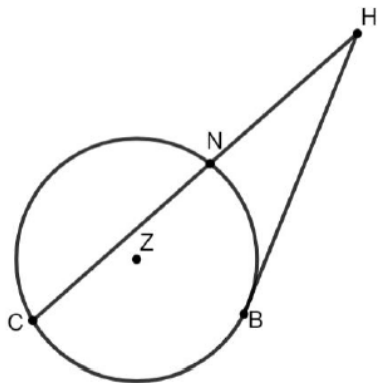
12.3.4 Your Turn!

Component Overview: Students are practicing finding lengths of intersecting secant and tangent and identifying errors made in work.



12.3.4 Your Turn!

1. Circle Z is shown with points C , N , and B on the circle. Secant $\overline{CH}$ and tangent $\overline{BH}$ intersect at point H outside the circle. $CN = (7x - 6)$, $NH = 12$, and $HB = 18$. Find the value of x .



$$18^2 = 12(7x - 6 + 12), \quad x = 3$$



Prompt students to label their diagrams. Students who do not label the diagram and try to solve the problem by just reading the instructions are likely to make mistakes.



Consider asking guiding questions to help struggling students:

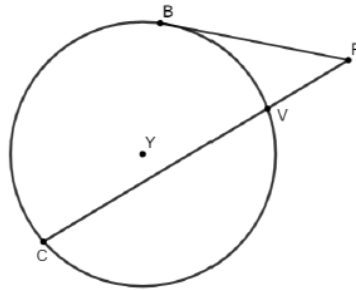
- Which line segment is the tangent?
- What do you know about the length of the tangent?
- Which line segment is the secant?
- What do you know about the length of the secant?



Consider challenging students to engage with question 1 or question 2 without using the image first. This will challenge students to use the information given to identify the error made.

12.3.4 Your Turn! (continued)

2. Isabella and Gregory were working on finding the length of $\overline{BP}$. Circle Y is given, where B , V , and C are points on the circle. Secant $\overline{CP}$ and tangent $\overline{BP}$ intersect at point P outside the circle. $CV = 12$ and $VP = 4$.



MA.K12.MTR.4.1: Engage in discussions that reflect on the mathematical thinking of self and others. Students gain valuable practice for metacognition and analyzing the work of others. Students identify the error in addition to thinking about Gregory's thinking in this work. For students who are unsure how to respond, use prompts like "What do you think Gregory was thinking when he did that? What should he do instead?"

Isabella's and Gregory's work to find the length of $\overline{BP}$ is shown.

Isabella's Work	Gregory's Work
$x^2 = 4(4 + 12)$	$x^2 = 12(4 + 12)$
$x^2 = 4(16)$	$x^2 = 12(16)$
$x^2 = 64$	$x^2 = 192$
$x = 8$	$x = 13.9$

- Whose work do you agree with?
Isabella
- Write a note to the person whose work is incorrect explaining their mistake.
Gregory, you need to find the product of the lengths of the secant segment and its external segment. You found the product of the secant segment and the internal segment.

12.3.5 Wrap-Up: Cool Down

Component Overview: Students write a short summary of their learning in the lesson. Student responses can be used to identify misconceptions and plan for upcoming instruction.



12.3.5 Wrap-Up: Cool Down

1. Write a summary of what you learned in this lesson.

Answers will vary.



What would reflection look like for you as a teacher? Consider your own scale for strengths and gaps for students through the lesson and how best to fill these gaps. Use this reflection to assess if your students need further clarification of this topic.